



BIRZEIT UNIVERSITY

Faculty of Engineering Technology

Electrical & Computer Engineering Department

ENCS5341, Machine Learning and Data Science

Project Report

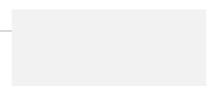
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Introduction

1.1 Task Description

This project addresses a multi-class text classification problem: predicting the country of a travel destination based on its textual description. Given a dataset of 1,011 travel destination samples collected from student submissions, the goal is to classify each destination into one of 54 different countries using only the text description field.

This task is challenging for several reasons:

- High number of classes (54 countries) makes it significantly harder than binary or few-class classification
- Class imbalance exists, with some countries having 60+ samples while others have fewer than 10
- Textual similarity between descriptions of geographically or culturally similar countries (e.g., Greece vs. Turkey, Japan vs. Korea)
- Variable description quality, ranging from detailed landmark descriptions to generic phrases

Additionally, we implemented a hierarchical classification approach as a creative extension, where countries are first grouped into 7 regions (Europe, Asia, Middle East, Americas, Africa, Oceania, Other), and classification proceeds in two stages: Region → Country.

1.2 Models Explored

We represent each description using TF-IDF features, then train and compare:

Baseline: k-Nearest Neighbors (k=1 and k=3) in TF-IDF space.

Model 1: Logistic Regression (tuned using different C values).

Model 2: Random Forest (tuned using different numbers of trees).

1.3 Evaluation Metrics

We used the following metrics to evaluate model performance:

- **Accuracy:** The proportion of correctly classified samples out of all test samples.

$$Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

- **Precision (Weighted):** The weighted average of precision across all classes, where precision measures how many predicted positives are actually correct.

$$Precision = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

- **Recall (Weighted):** The weighted average of recall across all classes, where recall measures how many actual positives are correctly identified.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

- **F1-Score (Weighted):** The harmonic mean of precision and recall, providing a balanced measure especially useful for imbalanced datasets.

$$\text{F1 - Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

We used weighted averaging for all metrics to account for class imbalance, giving more importance to classes with more samples.

Additionally, we performed 5-fold cross-validation on training data to verify model stability and ensure results are not due to a particular train-test split.

Exploratory Data Analysis

2.1 Data Collection and Cleaning

The dataset was compiled from 121 CSV files submitted by students, containing travel destination information. The data loading and cleaning process involved several key steps:

Data Loading:

- Successfully loaded 112 files (9 files failed due to encoding issues)
- Combined into a single dataset with 1,147 raw samples
- Handled multiple encoding formats (UTF-8, Latin-1, ISO-8859-1, CP1252)

Column Standardization:

- Unified inconsistent column names (e.g., "ImageURL", "image_url", "URL" → "Image URL")
- Standardized to 9 columns: Image URL, Description, Country, Weather, Time of Day, Season, Activity, Mood/Emotion, Student ID

Data Cleaning:

- Removed 95 duplicate rows based on Image URL and Description
- Standardized text values (e.g., "Usa" → "USA", "United Kingdom" → "UK")
- Corrected weather values containing invalid entries
- Final cleaned dataset: 1,011 samples

2.2 Dataset Overview

After cleaning, the dataset contains 1,011 samples and 9 columns. The target label for this project is Country, and the main input feature is the Description text.

We also checked the number of unique values (especially countries) and basic summary information to understand the data scale and structure.

2.3 Missing Values and Data Quality

Some columns contain missing values (e.g., Weather, Time of Day, Season, Activity, Mood/Emotion). Since our prediction task uses the Description text to predict Country, missing values in the other columns do not block the main model pipeline. However, these missing values are a data quality challenge and can limit using those fields as features without preprocessing (e.g., filling missing values).

2.4 Distributions and Interesting Trends

We visualized key distributions to understand the dataset:

- **Country distribution:** the dataset shows a long-tail imbalance (few countries have many samples while many countries have few). This may reduce performance for rare classes.
- **Weather / Season distribution:** these provide context about the dataset composition and how students described destinations.
- **Description length distribution:** description lengths vary widely, which supports using TF-IDF (handles variable-length text well). Very long descriptions may behave as outliers.

EXPLORATORY DATA ANALYSIS

Dataset Shape: (1011, 9)
Number of Countries: 121

Top 20 Countries:

Country	
Italy	66
Japan	63
France	53
Spain	52
Switzerland	52
USA	49
Palestine	45
Maldives	41
Turkey	40
Greece	39
Egypt	29
Saudi Arabia	25
UAE	25
Iceland	21
UK	21
China	20
Jordan	19
Canada	19
Brazil	17
Indonesia	16

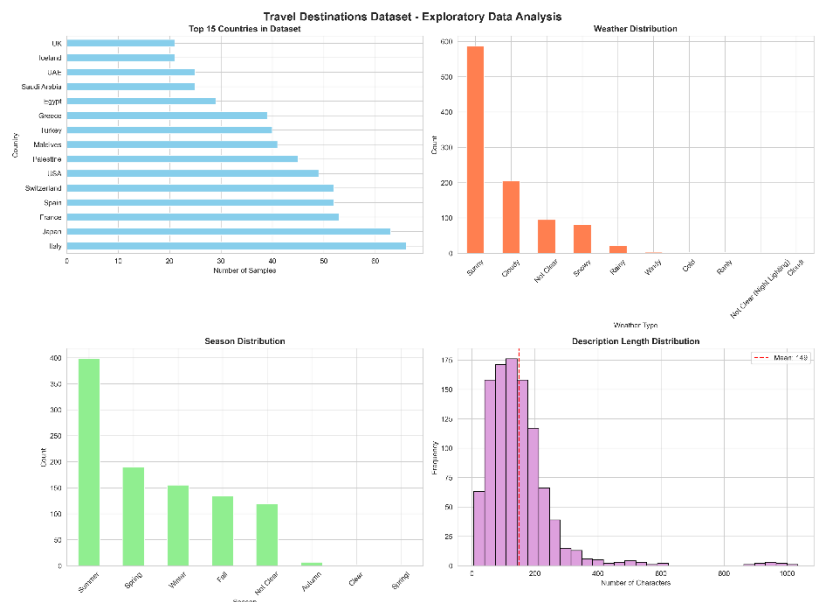
Name: count, dtype: int64

Missing Values:

Image URL	30
Description	0
Country	0
Weather	10
Time of Day	45
Season	6
Activity	26
Mood/Emotion	68
student_id	0
dtype:	int64

Description Length Statistics:

Min: 6
Max: 1034
Mean: 149.14



2.5 EDA Challenges Found

Main challenges discovered during EDA:

- **Class imbalance** (many countries with few samples).
- **Missing values** in several non-text columns.
- **High variation in description length** (short vs very long descriptions).

3. Experiments and Results

3.1 Experimental setup

- **Modeling dataset (after filtering):** we removed countries that have < 3 samples to reduce extreme imbalance and make evaluation more reliable. Final modeling set: 925 samples, 54 countries (classes).
- **Split:** stratified 80/20 train-test → 740 train, 185 test.
- **Text features:** TF-IDF on Description with:

max_features=1000, ngram_range=(1,2), stop_words='english', min_df=2, max_df=0.8.

```
After filtering (countries with ≥3 samples):
Total samples: 925
Number of countries: 54

Feature (X) shape: (925,)
Target (y) shape: (925,)
Target classes: 54

Train-Test Split (80-20):
Training samples: 740
Test samples: 185

Feature Extraction using TF-IDF...
TF-IDF feature shape: (740, 1000)
Vocabulary size: 1000
Sample features: ['activity', 'adding', 'adventure', 'adventurous', 'aegean', 'aegean sea', 'aerial', 'aerial view', 'africa', 'afternoon', 'air', 'air balloons', 'al', 'al aqsa', 'al haram', 'alhambra', 'alhambra palace', 'alive', 'alpine', 'alpine scenery']
```

3.2 Baseline model: k-Nearest Neighbors (k-NN)

We used k-NN as a simple baseline on TF-IDF vectors:

- **k = 1:** Accuracy **36.76%**, Weighted F1 **0.4056**
 - **k = 3:** Accuracy **18.92%**, Weighted F1 **0.1951**
- Best baseline is **k = 1**.

BASELINE MODEL: K-NEAREST NEIGHBORS

1. k-NN with k=1:
Accuracy: 0.3676 (36.76%)
F1-Score (weighted): 0.4056
2. k-NN with k=3:
Accuracy: 0.1892 (18.92%)
F1-Score (weighted): 0.1951

Better baseline: k=1

3.3 Proposed model 1: Logistic Regression (with tuning)

Logistic Regression is a strong baseline for text classification because it works well with high-dimensional sparse TF-IDF vectors and provides a linear decision boundary. We trained Logistic Regression on the TF-IDF features extracted from the destination descriptions and used it as our first proposed model.

To improve performance, we tuned the regularization parameter C.

- C controls the strength of regularization:
 - Small C → stronger regularization → simpler model (may underfit).
 - Large C → weaker regularization → more flexible model (may overfit).
- We evaluated several C values and selected the one that gives the best test performance (mainly accuracy and weighted F1). After choosing the best C, we report the final evaluation metrics and also check stability using 5-fold cross validation.

```
MODEL 1: LOGISTIC REGRESSION
Hyperparameter Tuning (C parameter):
Testing 4 different regularization strengths...

C= 0.01 → Accuracy: 0.0703 (7.03%), F1: 0.0092
C= 0.10 → Accuracy: 0.3297 (32.97%), F1: 0.2913
C= 1.00 → Accuracy: 0.5730 (57.30%), F1: 0.5024
C= 10.00 → Accuracy: 0.6649 (66.49%), F1: 0.6106
```

```
Best C parameter: 10.0
Best Accuracy: 0.6649 (66.49%)
```

```
Final Logistic Regression Performance:
Accuracy: 0.6649
Precision: 0.6071
Recall: 0.6649
F1-Score: 0.6106
```

Improvement over baseline: +80.88%

CROSS-VALIDATION RESULTS (Logistic Regression)

5-Fold Cross-Validation:

```
Mean Accuracy: 0.6662 (+/- 0.0397)
Individual Folds: ['0.6284', '0.6824', '0.6689', '0.6824', '0.6689']
This confirms the model is stable and not overfitting.
```

To better understand the Logistic Regression model, we inspected the most discriminative words (highest positive coefficients) for several frequent countries. These words act as strong signals in the description text and explain why the model can separate classes effectively. For example, *Italy* is strongly associated with words like “venice” and “rome”, while *France* is associated with “eiffel” and “paris”. This confirms that TF-IDF + Logistic Regression learns meaningful country-specific keywords from the dataset.

FEATURE IMPORTANCE: Top Discriminative Words per Country

Top 5 most predictive words for each major country:

Italy	: italy, venice, rome, canal, colosseum
Japan	: tokyo, japan, japanese, kyoto, torii
France	: eiffel tower, eiffel, paris, foreground, tower
Spain	: barcelona, sagrada, alhambra, palace, madrid
Switzerland	: swiss, mountains, switzerland, country, green

These words have the highest positive coefficients for each country, meaning they strongly indicate that country in the text description.

3.4 Proposed model 2: Random Forest (with tuning)

We trained a Random Forest classifier as a second proposed model using the same TF-IDF representation of the description text. Random Forest is an ensemble of decision trees and can model non-linear patterns, so we used it to compare against the linear Logistic Regression model.

Hyperparameter tuning

We tuned `n_estimators` (number of trees) and selected the value that achieved the best test performance.

From our experiments, the best configuration was `n_estimators = 100`, with the following test results. We also performed 5-fold cross-validation, obtaining mean accuracy 0.6176 :

MODEL 2: RANDOM FOREST

Hyperparameter Tuning (`n_estimators` parameter):
Testing 4 different numbers of trees...

```
n_estimators= 50 → Accuracy: 0.5784 (57.84%), F1: 0.5497
n_estimators=100 → Accuracy: 0.5838 (58.38%), F1: 0.5454
n_estimators=200 → Accuracy: 0.5838 (58.38%), F1: 0.5508
n_estimators=300 → Accuracy: 0.5838 (58.38%), F1: 0.5506
```

Best `n_estimators`: 100
Best Accuracy: 0.5838 (58.38%)

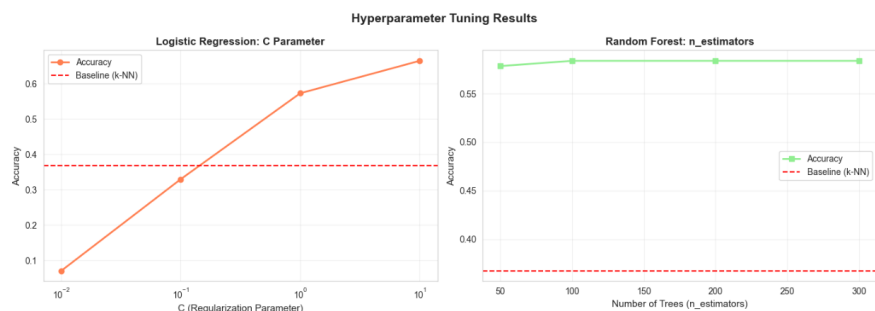
Final Random Forest Performance:
Accuracy: 0.5838
Precision: 0.6236
Recall: 0.5838
F1-Score: 0.5454

CROSS-VALIDATION RESULTS (Random Forest)

5-Fold Cross-Validation:

Mean Accuracy: 0.6176 (+/- 0.0457)

Individual Folds: ['0.5811', '0.6419', '0.6284', '0.6014', '0.6351']



3.5 Creative Approach: Hierarchical Classification (Region → Country)

Instead of predicting the country directly among 54 classes, we tested a two-stage hierarchical classifier. In Stage 1, we map each country to a broader region (e.g., Europe, Asia, Middle East) and train a Logistic Regression ($C=10$) model to predict the region from

the TF-IDF description features. In Stage 2, we train a separate Logistic Regression (C=10) classifier per region to predict the country only within the predicted region (with a fallback to the flat Logistic Regression if needed).

This approach improves interpretability because it explains decisions in two steps (region first, then country). On the test set, the hierarchical model achieved Accuracy = 0.6629 (66.29%) and Weighted F1 = 0.6115, which is very close to the flat Logistic Regression

HIERARCHICAL CLASSIFICATION - TWO-STAGE APPROACH

[STEP 1] Creating region mapping...
Mapped 54 countries to regions

Region distribution:
Region
Europe 419
Middle East 159
Asia 148
Americas 103
Oceania 55
Other 33
Africa 8
Name: count, dtype: int64

[STEP 2] Preparing hierarchical train-test split...
Training samples: 750
Test samples: 175

[STEP 3] Training Stage 1: Region Classifier...
Stage 1 (Region) Accuracy: 0.7829 (78.29%)

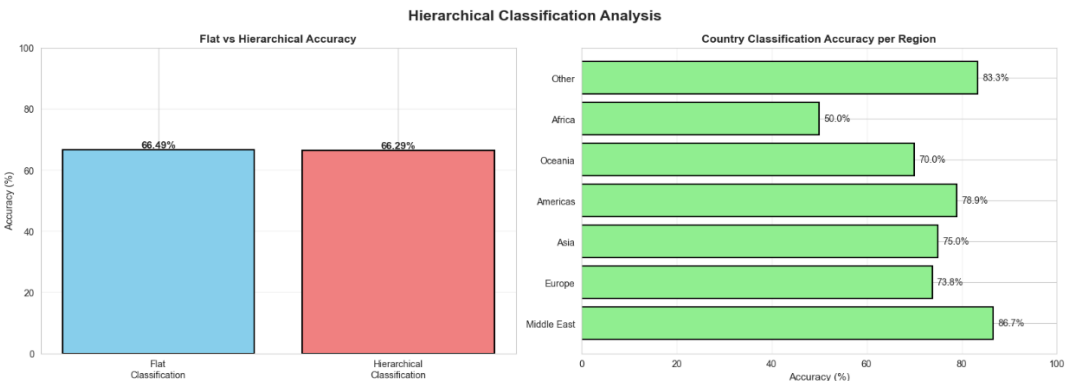
Region Classification Report:				
	precision	recall	f1-score	support
Africa	0.00	0.00	0.00	2
Americas	0.67	0.42	0.52	19
Asia	0.72	0.75	0.74	28
Europe	0.79	0.93	0.85	80
Middle East	0.86	0.83	0.85	30
Oceania	0.80	0.80	0.80	10
Other	1.00	0.17	0.29	6
accuracy			0.78	175
macro avg	0.69	0.56	0.58	175
weighted avg	0.78	0.78	0.76	175

[STEP 4] Training Stage 2: Country Classifiers per Region...
Middle East - Countries: 8, Test samples: 30, Accuracy: 0.8667
Europe - Countries: 16, Test samples: 80, Accuracy: 0.7375
Asia - Countries: 11, Test samples: 28, Accuracy: 0.7500
Americas - Countries: 5, Test samples: 19, Accuracy: 0.7895
Oceania - Countries: 3, Test samples: 10, Accuracy: 0.7000
Africa - Countries: 2, Test samples: 2, Accuracy: 0.5000
Other - Countries: 9, Test samples: 6, Accuracy: 0.8333

[STEP 5] Running full hierarchical pipeline...

Hierarchical Classification Results:
Accuracy: 0.6629 (66.29%)
F1-Score: 0.6115
COMPARISON: Flat vs Hierarchical Classification

Approach	Accuracy	F1-Score	Interpretability	Complexity
Flat (Direct Country)	0.664865	0.610647	Low	Simple
Hierarchical (Region→Country)	0.662857	0.611489	High	Complex



3.6 Model Comparison

The final test performance of the main models. The best overall model is Logistic Regression (C=10) with Accuracy = 0.6649 and Weighted F1 = 0.6106. Random Forest achieved Accuracy = 0.5838 and Weighted F1 = 0.5454, while the k-NN baseline was much lower (best baseline k=1 with Accuracy = 0.3676). The hierarchical approach achieved similar performance to the best flat model (66.29% vs 66.49%) while offering higher interpretability.

ENHANCED MODEL COMPARISON SUMMARY

	Model	Accuracy	Precision	Recall	F1-Score
	k-NN (k=1)	0.367568	0.623912	0.367568	0.405581
	k-NN (k=3)	0.189189	0.407541	0.189189	0.195056
	Logistic Regression (C=10.0)	0.664865	0.607057	0.664865	0.610647
	Random Forest (n_est=100)	0.583784	0.623553	0.583784	0.545405

3.7 Error Analysis

We analyzed errors made by the best model on the test set. The model produced 62 errors out of 185 test samples, giving an error rate of 33.51%. The confusion matrix (top 10 most frequent countries) shows that some classes are much easier than others. For example, per-class accuracy is high for Spain, Switzerland, Palestine, Maldives (100%), but much lower for USA (40%).

The most common confusion pairs included USA → Switzerland (2), Turkey → Italy (2), USA → Italy (2), Australia → Maldives (2), and Japan → USA (2), which together account for 25.8% of all errors. We also checked whether text length affects performance: accuracy was similar for short (<100) and medium (100–200) descriptions (~0.66–0.68), and slightly lower for long (200–500) descriptions (~0.64).

ERROR ANALYSIS

Total errors: 62 out of 185 test samples
Error rate: 33.51%

4. Analysis

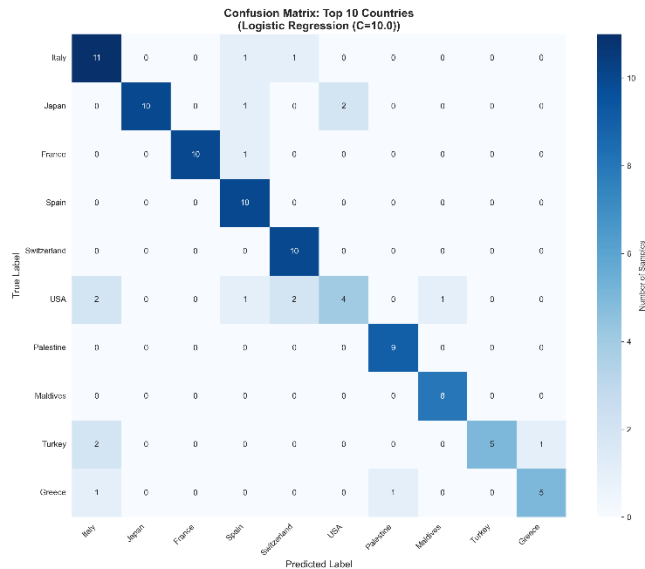
4.1 How we analyzed model errors

To analyze errors of the best model (Logistic Regression, C=10), we compared the predicted labels with the true labels on the **test set**. We then:

1. counted the total number of wrong predictions (errors) and the error rate,
2. built a confusion matrix for the top 10 most frequent countries (to focus on the main classes),
3. extracted the most common misclassification pairs (True → Predicted), and
4. checked whether error rate changes with description length by grouping test samples into length ranges.

4.2 Confusion matrix insights (Top 10 countries)

The figure shows the confusion matrix for the top 10 countries using the best model (Logistic Regression, C=10). Most predictions fall on the diagonal, but repeated confusions appear between specific country pairs (e.g., USA→Switzerland and Turkey→Italy). The top 10 confusion pairs account for 16/62 (25.8%) of all errors, which means a small set of mistakes explains many wrong predictions. Accuracy is similar for short and medium descriptions and slightly lower for longer ones, so errors are mainly caused by overlapping keywords and class imbalance rather than text length.



MOST CONFUSED COUNTRY PAIRS

Top 10 Most Common Misclassification Pairs:

True Country	→ Predicted	: Count
USA	→ Switzerland	: 2
Turkey	→ Italy	: 2
USA	→ Italy	: 2
Australia	→ Maldives	: 2
Japan	→ USA	: 2
Thailand	→ Maldives	: 2
Italy	→ Spain	: 1
New Zealand	→ USA	: 1
Korea	→ Japan	: 1
Saudi Arabia	→ Palestine	: 1

These top 10 pairs account for 16/62 (25.8%) of all errors

PER-CLASS ACCURACY (Top 10 Countries)

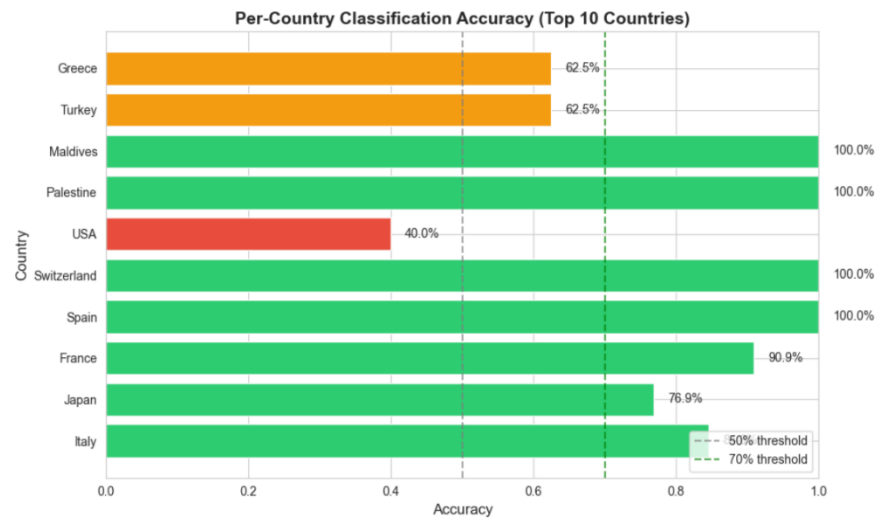
Country	Test Samples	Correct	Accuracy
Italy	13	11	84.62%
Japan	13	10	76.92%
France	11	10	90.91%
Spain	10	10	100.00%
Switzerland	10	10	100.00%
USA	10	4	40.00%
Palestine	9	9	100.00%
Maldives	8	8	100.00%
Turkey	8	5	62.50%
Greece	8	5	62.50%

Easiest to classify: Spain (100.00%)

Hardest to classify: USA (40.00%)

Accuracy by Description Length Category:

	Accuracy	Sample Count
length_category		
Short (<100)	0.661538	65
Medium (100-200)	0.678571	84
Long (200-500)	0.638889	36
Very Long (>500)	NaN	0



5. Conclusions and Discussion

This project addressed multi-class text classification: predicting the country of a travel destination from its description using TF-IDF features. Among the tested models, Logistic Regression ($C=10$) achieved the best overall performance (Accuracy = 0.6649, Weighted F1 = 0.6106), outperforming Random Forest (Accuracy = 0.5838) and the k-NN baseline. The hierarchical (Region $\rightarrow$ Country) creative approach achieved very similar performance (Accuracy = 0.6629, Weighted F1 = 0.6115), showing that adding structure can improve interpretability without sacrificing accuracy.

Limitations: The dataset has class imbalance (many countries have few samples), which reduces performance for rare classes and makes the evaluation depend strongly on the most frequent countries. Some columns also contain missing values, and TF-IDF ignores deeper semantic meaning (it relies mainly on keyword overlap). In addition, accuracy and weighted F1 summarize overall performance but may hide poor performance on minority classes.

Future improvements: Performance could be improved by

- applying class weighting or re-sampling to reduce imbalance.
- using stronger text representations such as word embeddings or transformer models.
- expanding the hierarchical idea with better region definitions and a confidence-based fallback.
- reporting macro-F1 to better reflect performance on rare classes.